



if I have a logistic depot with a PV and I have battery electric trucks. should I calculate charging my bev trucks from my pv with the typical value of PV generated energy of around 7c/kWh or with the opportunity cost because I could also sell the generated PV energy at the spot market for lets say 10c/kWh. how sould I price the bev truck charging costs from my own PV?

For the economic cost of charging an BEV truck with PV electricity that could otherwise be exported, use the opportunity cost of the PV energy—i.e., the *net revenue you forgo by not selling it*—not simply the PV system’s average generation cost of 7 ct/kWh.

So, if 1 kWh of PV electricity can be exported at 10 ct/kWh at that moment, allocating it to truck charging carries an energy cost of roughly 10 ct per kWh at the PV meter (adjusted for charging/conversion losses). The 7 ct/kWh figure is useful for long-run investment appraisal and cost allocation, but it is usually not the right marginal price for an operational charging decision.

The key distinction

Question	Appropriate PV value	Why
“Should the truck charge now rather than export PV?”	Opportunity cost / export value	Charging consumes a kWh that could have earned export revenue
“What does the PV system cost over its lifetime?”	LCOE / average PV generation cost , e.g. 7 ct/kWh	Includes CAPEX, financing, O&M, degradation, lifetime yield
“What is the internal transfer price from PV asset to fleet?”	Usually time-varying opportunity cost , potentially plus losses and tariffs	Gives the fleet the economically correct incentive
“Was installing more PV or storage worthwhile?”	Full-system NPV / LCOE / marginal investment cost	This is a long-term capacity-investment decision

Behind-the-meter PV is normally consumed on site first, while excess production is exported and remunerated at a sell rate. Depending on the tariff, that export compensation may be a fixed feed-in tariff, net-billing sell rate, or time-varying wholesale/spot value. ^[1] ^[2]

Recommended charging-cost formulation

For each time interval t , calculate the cost of PV-origin electricity delivered to the truck as:

$$\left[\frac{p_{\mathrm{export},t} + c_{\mathrm{var,PV},t} + c_{\mathrm{additional},t}}{\eta_{\mathrm{PV\to truck},t}} \right]$$

Where:

- $p_{\mathrm{export},t}$: **net export revenue forgone** in ct/kWh of PV AC energy. Use the actual settlement value—not necessarily the headline spot price.
- $c_{\mathrm{var,PV},t}$: incremental variable cost, generally small; potentially inverter wear, metering/administrative costs, or contract-specific charges.
- $c_{\mathrm{additional},t}$: charging-specific costs not covered elsewhere, such as a bidirectional converter, stationary-battery degradation, or a self-consumption levy if applicable.
- $\eta_{\mathrm{PV\to truck},t}$: total efficiency from PV meter to truck battery, including transformer, cabling, AC/DC conversion, and battery charging losses.

If the 10 ct/kWh is a genuine net export value and total PV-to-battery efficiency is 92%, then:

$$c_{\mathrm{PV\to battery}} = \frac{10}{0.92} = 10.87 \text{ ct/kWh}_{\mathrm{battery}}$$

Thus, 1 kWh actually stored in the truck battery should be priced at about **10.9 ct**, before any additional charger depreciation, demand-charge effects, or taxes.

When 7 ct/kWh is appropriate

Your 7 ct/kWh is presumably the PV system's **levelised cost of electricity**. It should be used in a long-term model, for example when comparing:

- PV capacity expansion versus buying electricity from the grid,
- PV-direct charging versus a dedicated PPA,
- PV plus stationary storage versus PV-only charging,
- the total cost of ownership of the depot-fleet energy system.

But do not treat it as a saving of $p_{\mathrm{grid}} - 7$ ct/kWh if that PV kWh has an alternative export value of 10 ct/kWh. Doing so double-counts the economic benefit: the company either earns 10 ct by exporting it or avoids buying a grid kWh by using it internally. The economically relevant gain from self-consuming that marginal kWh is:

[$\mathrm{Value\ of\ PV\ self\ consumption}_t$

$p_{\mathrm{grid,avoided},t}$

$p_{\mathrm{export},t}$

]

not:

$$p_{\mathrm{grid,avoided},t} - \mathrm{LCOE}_{\mathrm{PV}}$$

For example, with a 25 ct/kWh marginal grid-purchase price and a 10 ct/kWh export value, charging with PV instead of exporting creates a net operating benefit of roughly 15 ct/kWh of PV electricity—before losses and demand-charge effects. Export remuneration is commonly below the retail import rate under net-billing-style arrangements, which is why self-consumption is often financially attractive. ^[3] ^[4]

Practical depot methodology

For a technically robust truck-depot model, I would report **three separate metrics**, rather than force a single PV cost number onto all purposes:

1. Physical PV share

- Direct PV-to-truck energy, PV-via-stationary-storage energy, grid-to-truck energy.
- Allocate PV only when PV generation and charging overlap physically, unless you explicitly model a battery or a contractual virtual allocation.

2. Marginal operational charging cost

- For direct PV charging: net foregone export revenue, time-resolved.
- For grid charging: marginal import price, including energy tariff, relevant levies/taxes, balancing costs, and—where applicable—incremental demand charges.
- For PV charged first into a stationary battery: PV opportunity cost plus stationary-battery round-trip losses and degradation.

3. Long-run economic cost

- Annualised PV CAPEX, O&M, financing, replacement costs, degradation, charger CAPEX, stationary storage CAPEX, and grid-connection costs.
- Deduct export revenues as a project cash flow.
- Divide by truck-battery energy delivered, vehicle-km, tonne-km, or annual fleet service—depending on your research question.

Important exceptions

Use a value below 10 ct/kWh only if 10 ct is not actually revenue you would receive:

- The 10 ct is an unadjusted wholesale spot price, but your settlement is spot price minus balancing, market-access, aggregation, or other charges.
- Export is capped, unavailable, or curtailed; then the alternative value may be near zero during those intervals.
- The site has an annual net-metering arrangement in which exported kWh offset later imports at a different value; then model the actual billing rules and settlement period rather than hourly spot prices.
- The PV plant is subject to a fixed feed-in contract; use the contractual feed-in remuneration, not the spot price.
- Truck charging reduces a very expensive import demand peak. Then the avoided power/demand charge can be material and should be credited to the charging strategy separately from the energy price.

In the special case of otherwise-curtailed PV, the short-run opportunity cost may be close to zero—although the PV's LCOE remains 7 ct/kWh for long-run asset economics. PV export compensation can indeed range from market value or spot-price-based remuneration to a fixed tariff, retail credit, or no compensation, depending on the regulatory and contractual setup.^[5]
^[6]

Bottom line

For your example, the cleanest internal pricing rule is:

Price direct PV charging at the time-specific net export value that the depot gives up, divided by the PV-to-battery efficiency.

With 10 ct/kWh export value and 92% delivery efficiency, that is approximately 10.9 ct/kWh delivered to the truck battery. Keep the 7 ct/kWh PV LCOE as a separate long-term system-cost metric, not as the marginal operational charging cost.

This separation is particularly useful in a scientific depot/fleet model: it prevents PV self-consumption, export revenues, and avoided grid purchases from being conflated, while still allowing you to evaluate both dispatch and infrastructure-investment decisions consistently.

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